

Luis Victoria

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Work Experience

Software Engineer — Market Data Platform, Site Reliability & Performance

Feb 2023 - Jul 2026

Bloomberg LP

New York, NY

Worked on performance, scalability, and reliability across Bloomberg's Market Data Platform, spanning Feeds and the Ticker Plant. The platform handled >700B ticks/day during extreme volatility while maintaining <6ms median and <250ms p99.9 latency SLAs.

- **Predictive Capacity Forecasting:** Rewrote a fragile legacy log-scraping storage monitor as a two-part Python system: a fleet-wide scanner writing structured database and machine metadata to a centralized datastore, and a forecasting service extrapolating rolling usage trends 14 days ahead with automated alerting. Delivered in one week during the April 2025 tariff-driven volatility spike, protecting 1,000+ machines from storage-exhaustion/data-loss risk.
- **Performance Regression Detection:** Applied [Bayesian Online Changepoint Detection](#) within Argo-orchestrated workflows to statistically flag anomalous times across 30,000+ tick processor instances, cutting remediation time from days to <24 hours.
- **Latency Instrumentation Overhaul:** Authored a C++20 envelope-based time-stamping framework tracing tick updates stage-by-stage through the Ticker Plant pipeline with μ s resolution, enabling latency regression detection and optimization with < 10 μ s overhead.
- **Disaster Recovery Performance Resilience:** Automated load reassignment to/from the cluster's real-time broadcast ('monitor') machine during data center failover, removing manual on-call intervention. Benchmarked CPU thread-pinning under combined load, informing Linux cgroups-based CPU prioritization to preserve broadcast continuity while controlling dropped requests.
- **Capacity & Load Management:** Assessed CPU, memory, and storage capacity ahead of onboarding new global exchanges and executed targeted cluster splits to offload overloaded machines, avoiding or eliminating SLO breaches with minimal client disruption.

Quantitative Developer Intern

May 2019 - Aug 2019

Farrington Asset Management

Singapore

- **Back-testing Enhancements:** Extended the firm's Python-based back-testing engine to include cryptocurrency asset analysis by integrating new data pipelines using CCXT and Binance, enabling quantitative evaluation of crypto markets.
- **Automation with Bloomberg API:** Reduced time to generate client portfolio analysis by 50% for the portfolio management team using information from PORT<GO>.
- **Technical Research:** Leveraged Bollinger Bands and Stochastic Oscillators to support quantitative decision-making on biomedical pharmaceutical startups.

Education

Business and Computer Science

Sep 2018 - Dec 2022

Bachelor of Commerce - Sauder School of Business, University of British Columbia

Vancouver, BC

- **Relevant Coursework:** Machine Learning & Data Mining, Advanced Algorithmic Design, Advanced Relational Databases, Theory of Computation, Functional & Logic Programming, Matrix Algebra, Empirical Economics, Calculus III.
- **Sauder Trading Simulation (1st of 136 students):** Built a Raspberry Pi bot to poll a primitive, non-matching order-book trading platform every 2 minutes, detecting and executing on crossed bid/ask orders to run an automated arbitrage, outperforming classmates' fundamentals-based strategies through market-structure exploitation alone.